Title: Virtual Dine Restaurant Ordering Hub

Student Name: Sonu Paswan
Student ID: 2520090204
Cluster:01
Section: 01

Under the Guidance
of
Supervisor Name: Ramakanth
Assistant Professor
Department of CSE

Abstract

- Virtual Dine – Restaurant Ordering Hub is a **web-based restaurant ordering system** designed to simplify the dining experience using a digital interface.
- The application allows users to:
- Select different **dining rooms** such as Couple Room, Family Room, or Regular Table.
- Browse food items based on **multiple cuisines**.
- Add food items to a **shopping cart dynamically**.
- The system is developed using **HTML, CSS, and JavaScript** and provides a **responsive and interactive user interface**. Cart data is stored using **Local Storage**, allowing temporary storage of selected items.
- This project demonstrates the implementation of **dynamic UI updates, DOM manipulation, and event handling** in frontend development.

Existing System

- In traditional restaurant systems:
- Customers need to **physically visit the restaurant** to place orders.
- Menu browsing is done **manually using printed menus**.
- There is **no digital room selection system**.
- Food ordering requires **waiter interaction**.
- Order management is **not automated**.
- These limitations make the process slower and less efficient for both customers and restaurant staff.

Proposed System

- The **Virtual Dine system** provides a modern solution for restaurant ordering.
- Features include:
- Digital **room selection system** (Couple Room, Family Room, Regular Table).
- **Multiple cuisine filtering** such as Italian, Indian, Chinese, etc.
- **Dynamic food menu display**.
- Users can **add or remove items from cart**.
- **Automatic total price calculation**.
- Simple and **interactive user interface**.
- This system improves the **user experience and ordering efficiency**.

System Architecture

- The system follows a **client-side web architecture**.
- Main components:
- **Frontend Layer**
- HTML – Structure of the webpage
- CSS – Styling and layout design
- **Application Logic Layer**
- JavaScript handles:
  - Room selection
  - Cuisine filtering
  - Cart management
  - Order placement
- **Data Handling**
- Food items stored using **JavaScript arrays and objects**
- Cart data stored using **Local Storage**

Project Modules

- The project is divided into the following modules:
- **1. Room Selection Module**
- Allows users to select:
- Couple Room
- Family Room
- Regular Table
- **2. Cuisine Filtering Module**
- Users can filter food based on cuisine types:
- Italian
- Indian
- Chinese
- Mexican
- Japanese
- Thai
- Mediterranean
- Indo-Chinese

- **3. Food Display Module**
- Displays food cards containing:
- Food Image
- Food Name
- Price
- Add to Cart button
- **4. Cart Management Module**
- Handles:
- Adding items to cart
- Removing items from cart
- Total price calculation
- **5. Order Placement Module**
- Allows users to confirm and place the final order.

Objectives

- The main objectives of the project are:
- To design a **digital restaurant ordering system**.
- To implement **room selection functionality**.
- To allow users to **browse multiple cuisines**.
- To develop a **dynamic food menu interface**.
- To implement **cart management using JavaScript**.
- To create an **interactive and responsive UI**.

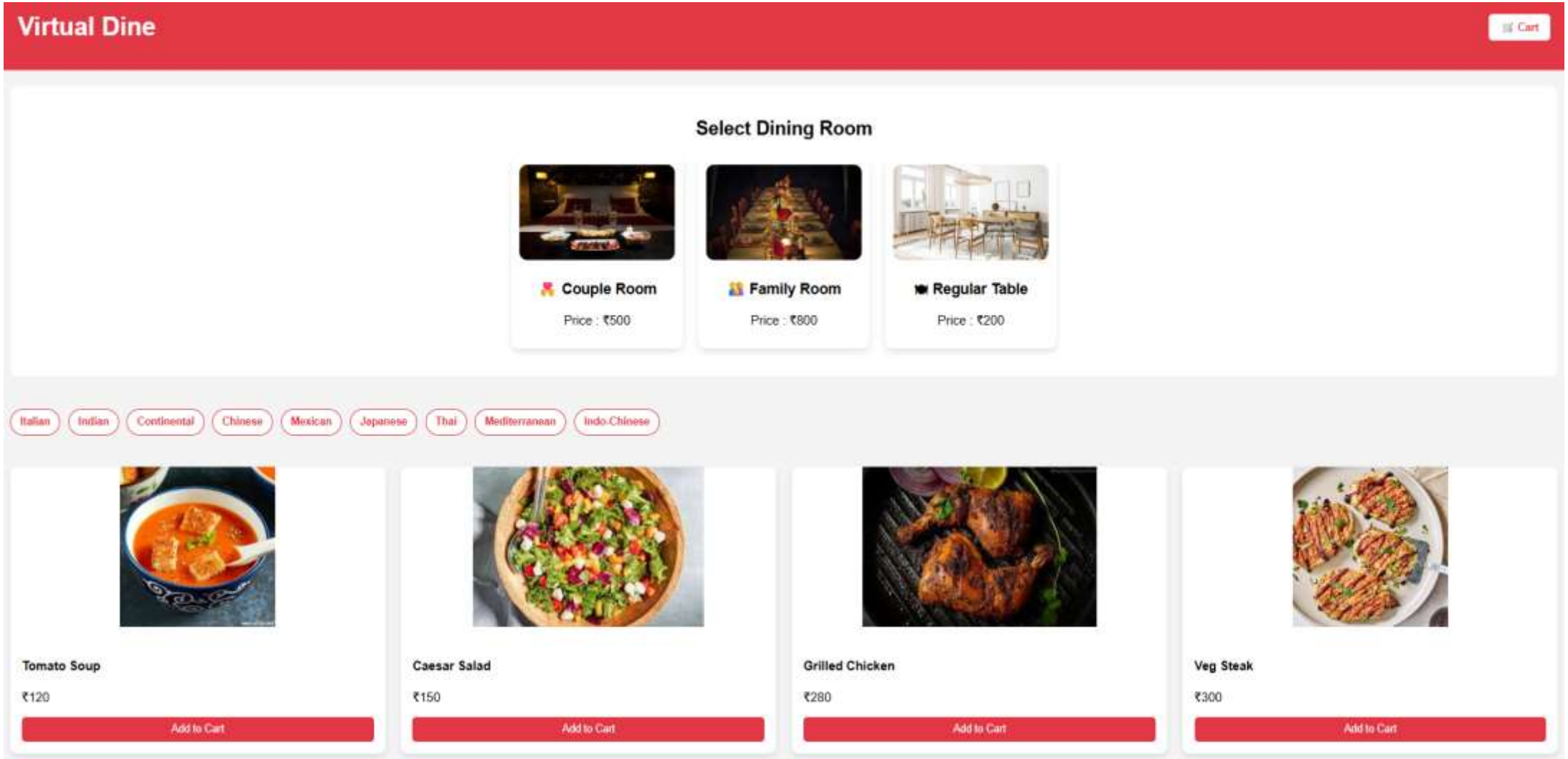
Co’s Attainments - FWD

- **CO1 – HTML Implementation**
- Use semantic HTML elements.
- Create structured webpages.
- **CO2 – CSS Styling**
- Apply layouts and responsive design.
- Style buttons, cards, and UI components.
- **CO3 – JavaScript Programming**
- Use arrays and objects to store food data.
- Implement logical conditions and functions.
- **CO4 – DOM Manipulation**
- Dynamically update webpage content.
- Handle user events like clicks and selections.

Co’s Attainments - DSA

- **CO1 – Basic Data Structures**
- Understand the concepts of basic data structures such as **arrays, linked lists, stacks, and queues** and their applications in solving problems.
- **CO2 – Algorithm Design**
- Develop algorithms to solve computational problems and understand the **step-by-step procedure** for problem solving.
- **CO3 – Complexity Analysis**
- Analyze the **time complexity and space complexity** of algorithms using **Big-O notation** to determine efficiency.
- **CO4 – Implementation of Data Structures**
- Implement various data structures such as **stacks, queues, and linked lists** and perform operations like **insertion, deletion, and traversal**.
- **CO5 – Sorting and Searching Techniques**
- Apply different **sorting algorithms** (Bubble Sort, Selection Sort, Insertion Sort) and **searching techniques** (Linear Search, Binary Search) for efficient data processing.
- **CO6 – Problem Solving Using Data Structures**
- Apply suitable **data structures and algorithms** to solve real-world problems effectively and optimize program performance.

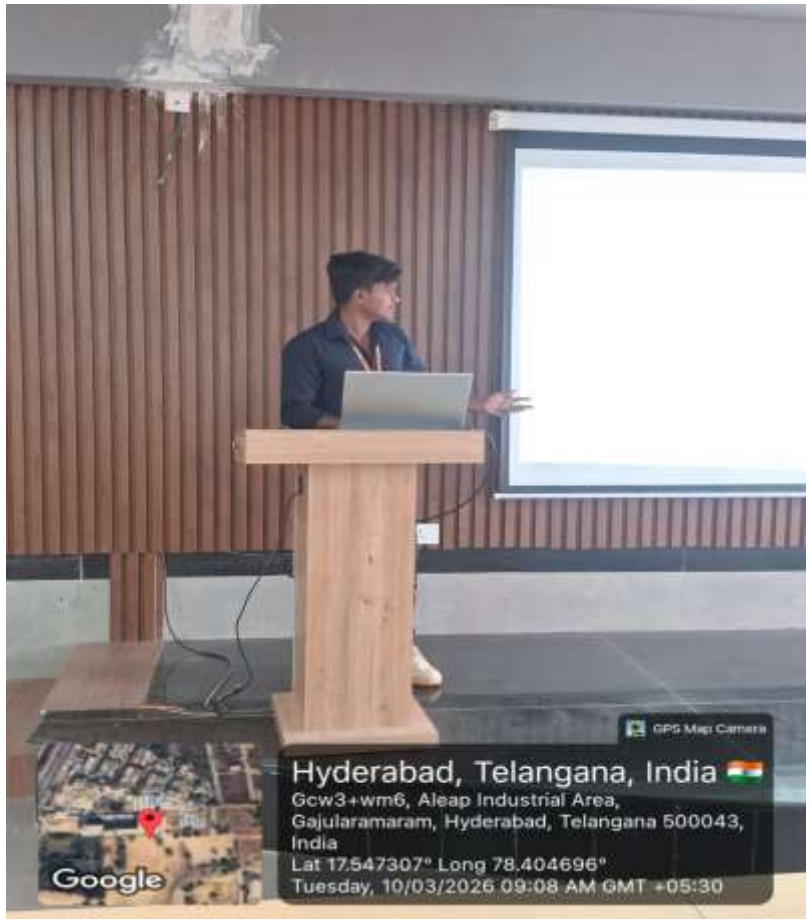
Screenshots for FWD & DSA



Conclusion

- Virtual Dine provides a **modern restaurant ordering solution** using web technologies.
- Key achievements of the project:
- Interactive and user-friendly interface.
- Dynamic food menu system.
- Real-time cart management.
- Room selection feature.
- Efficient order placement process.
- The project demonstrates practical knowledge of **HTML, CSS, JavaScript, DOM manipulation, and frontend development concepts**.

Geo- Tag Photos



THANK YOU